25900344 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) Regular EXAMINATION October 2025

Third SEMESTER

Discipline Specific Elective (DSE) - MG3DSEECO200 - INTRODUCTION TO DATA ANALYSIS FOR ECONOMICS

(2024 ADMISSION ONWARDS)

Duration: 1 Hour 30 Minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Multiple Choice Question

Answer all questions

Each question carries **1** marks

1. If interest is computed weekly, how many compounding periods are in one year? [A] / []

   a). 12 b). 52

   c). 365 d). 4

2. Which chart type is not ideal when there are more than 5–6 categories? [A] / []

   a). Column Chart b). Pie Chart

   c). Bar Chart d). Scatter Plot

3. Pivot Tables are best used for which type of data? [K] / []

   a). Text only documents b). Tabular data with rows and columns

   c). Small images d). Audio files

4. Which of the following is the BEST real-world example of Annuities? [An] / []

a). Buying a single stock share.

b). Depositing $200 into a savings account every month for five years.

c). Winning a lottery jackpot paid immediately.

d). Taking out a mortgage with variable monthly payments.

5. The coefficient of determination (R^2) is equal to: [K] / [CO1, CO2, CO3, CO4]

a). The square root of the correlation coefficient

b). The square of the correlation coefficient

c). The inverse of the correlation coefficient

d). The twice of the correlation coefficient

6. In spreadsheets, rows are identified by? [K] / [CO1, CO2, CO3, CO4]

a). Numbers

b). Letters

c). Symbols

d). Colors

7. The main purpose of VLOOKUP function in Excel is: [U] / []

a). Look for data horizontally in a table

b). Look for data vertically in a table

c). Filter a table

d). Sort a list

8. A researcher records the brands of smartphones owned by students. What type of data is this? [U] / []

a). Continous

b). Ratio

c). Nominal

d). Ordinal

9. A dataset yields a correlation coefficient of r = 0.8 between two variables. Apply your understanding to determine the correct interpretation of this value. [A] / []

a). The variables exhibit a strong positive linear relationship

b). The variables exhibit a strong negative linear relationship

c). The variables have no linear association

d). The variables are perfectly correlated

10. In regression, the coefficient "b" is called [U] / []

a). Correlation coefficient

b). Intercept

c). Variance

d). Regression Coefficient

11. To find the position of a character in a string (case-sensitive), we use: [U] / [CO1, CO2, CO3, CO4]

a). SEARCH() b). FIND()

c). MID() d). LEN()

12. Which of the following is true about sorting by cell color or font color in Excel? [A] / [CO1, CO2, CO3, CO4]

a). It is only available in Excel 2013 and earlier b). It can only be applied to numbers

c). It is an option available in the Custom Sort dialog d). It requires conditional formatting

13. Formula to find variance of a sample in excel is: [U] / []

a). =VAR.P() b). =VAR.S()

c). =STDEV.S() d). sd.p()

14. Cross-tabulation is mainly used to: [K] / []

a). Calculate mean and standard deviation b). Test hypothesis using regression

c). Summarize the relationship between two categorical variables d). Plot scatter diagrams

15. Command to find standard deviation of a population in excel is: [K] / []

a). sd.s() b). =STDEV.P()

c). stdev.s() d). sd.p()

[1x15 = 15]

Part B

Short Answer

Answer any **10** questions

Each question carries **2** marks

16. What does the CHAR() function return? Give an example with CHAR(65). [K] / [CO1, CO2, CO3, CO4]

17. Give the logical function command for the following problem. The bonus of a firm is calculated on the basis of sales. If sales are above Rs 10000/-, the bonus is 10% of sales. Otherwise the bonus is 0. [K] / []

18. Define Discrete Data. [U] / []

19. Differentiate between simple interest and compound interest with examples. [An] / []

20. What are Add-ins in data analysis tools? Provide examples. [K] / []

21. How to create a named cell? [A] / [CO1, CO2, CO3, CO4]

22. Differentiate between a Column Chart and a Bar Chart. [A] / []

23. Explain the procedure to calculate coefficient of variation Excel. [A] / []

24. What is non linear regression model? [K] / []

25. Mention two advantages of using spreadsheets in data analysis. [U] / [CO1, CO2, CO3, CO4]

26. Which function is used to count the number of numeric entries in a given range? Give an example. [U] / [CO1, CO2, CO3, CO4]

27. What is Cost Function? [K] / []

28. State the procedure for calculating mean in Excel? [A] / []

29. What is a key limitation of using IRR for investment decisions? [U] / []

[2x10 = 20]

Part C

Short Essay

Answer any **3** questions

Each question carries **5** marks

30. “Explain the concept and importance of a Scattergram (XY Scatter Plot) in data analysis. How does the CORREL() function in Excel complement the interpretation of a scattergram? [U] / []

31. Explain the difference between FIND() and SEARCH() with examples. [K] / [CO1, CO2, CO3, CO4]

32. What does the notation =A5+D$2 means while modelling algebraic models. Give the detailed application of the above command. [K] / []

33. The Weekly wages of workers in a factory are as follows: [U] / []

Workers	Wage (₹)
W1	400
W2	450
W3	500
W4	550
W5	600
W6	650
W7	1200

1. What is the mean wage of the workers?
2. What is the median wage?
3. If W7 is excluded, how do the mean and median change?
4. Calculate the total wage bill for all 7 workers.
5. Suppose management decides to set a “Standard Wage” equal to the median. How many workers earn below and above this wage.

34. If you expect to receive Rs 25000 in 5 years and the discount rate is 10%, what is the Present Value? [A] / []

35. From the below expressions,identify the operator (Arithemetic/Logical/Comparison) and write the results. [U] / []

Expression	Operators	Results
=30< 35		
= 5*7		
=91/8		
=9-13		
=50>=50		
=40<=35		
=AND(10>5,3<4)		
=OR(8>10,6<5)		
=90<>90		
=100=100		

36. Explain how IRR is used as a tool for evaluating mutually exclusive projects. Discuss the multiple IRR problem with an example [K] / []

[5x3 = 15]